



**wellcome
connecting
science**

CRISPR Screening and Analysis - Genome Informatics and Computational Design and Analysis

28 July - 1 August 2025, virtual via Zoom



**UNIVERSITY OF
WESTMINSTER**

PRE-COURSE ON LMS - Linux and R (for interest only)	Time	Monday 28 July	Tuesday 29 July	Wednesday 30 July	Thursday 31 July	Friday 1 August	Time				
	08:45	Join Call	Join Call	Join Call	Join Call	Join Call	08:45				
	09:00	Course introduction session Wellcome Connecting Science, training team, participants	Daily recap	Daily recap	Daily recap	Group project reminder Matthew Coelho and Kalpana Surendranath	09:00				
	09:15						09:15				
	09:30		Genome engineering workflows and screening Kalpana Surendranath	Applications of base editing and prime editing in pooled and arrayed screens Matthew Coelho	ENCORE Eloise Lockyer	Group work time	09:30				
	09:45									09:45	
	10:00									10:00	
	10:15									10:15	
	10:30	Break	Break	Break (download data for next session)	Break	Break	10:30				
	10:45						10:45				
	11:00	Introduction to the CRISPR toolbox Andrew Bassett	Case studies Munuse Savash Kalpana Surendranath	Base editing guide RNA design Cansu Dincer	CRISPR screening informatics Jamie Billington Saamin Cheema	Group work time	11:00				
	11:15									11:15	
	11:30	Guided computational practical: genes and genomes Saroor Patel								11:30	
	11:45									11:45	
	12:00									12:00	
	12:15						12:15				
	12:30						12:30				
	12:45						12:45				
	13:00	Lunch	Lunch	Lunch	Lunch	Lunch	13:00				
	13:15										13:15
	13:30										13:30
	13:45										13:45
	14:00	Refresher quiz Saroor Patel	CRISPResso and validation of gene-edited models Khalid Akram	SNV and knock-in validation (Sanger sequencing - TIDE) Sandeep Rajan	Seminar Manuel Kaulich	Group presentations: Groups of 6 10 min presentation/ group and 5min QA	14:00				
	14:15								14:15		
	14:30	Seminar Alice Refermat							14:30		
	14:45								14:45		
	15:00						Refresher quiz Jamie and Saamin		15:00		
	15:15						15:15				
	15:30	Intro to group project Matthew Coelho, Kalpana Surendranath	Refresher quiz Kalpana and Khalid	Refresher quiz Cansu and Sandeep	Project and practical Q&A	Wrap up and close	15:30				
	15:45										15:45
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